PCA in A Matrix Format

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PCA IN THE MATRIX FORMAT

Maximization of variance

- PCA maximizes the variance of the leading M principal components

$$\max_{\boldsymbol{b}_j} \boldsymbol{b}_j^\top \boldsymbol{S} \boldsymbol{b}_j \qquad \text{for } j = 1, \cdots, M$$

$$\text{subject to } \boldsymbol{b}_j^\top \boldsymbol{b}_k = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{if } j \neq k. \end{cases} \tag{1}$$

- We see that

$$\text{tr}(\boldsymbol{B}^\top \boldsymbol{S} \boldsymbol{B}) = \sum_{j=1}^{M} \boldsymbol{b}_j^\top \boldsymbol{S} \boldsymbol{b}_j$$

if $\boldsymbol{B} = (\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M) \in \mathbb{R}^{D \times M}$.

- We can then rewrite Eq. (1) in a matrix format as

$$\max_{\boldsymbol{B}} \text{tr}(\boldsymbol{B}^\top \boldsymbol{S}\boldsymbol{B})$$

$$\text{subject to } \boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_{M\times M}$$

Minimization of reconstruction error

- Alternatively, PCA can be formulated as minimization of the reconstruction error

$$\min_{\boldsymbol{B}} \sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}_n\|^2 \quad \text{subject to } \boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_{M\times M} \tag{2}$$

where $\boldsymbol{B} = (\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M)$ spans a M dimensional subspace.

Minimization of reconstruction error

- We can show that for $\boldsymbol{X} = (\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N)^\top \in \mathbb{R}^{N \times D}$

$$\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}_n\|^2 = \text{tr}((\boldsymbol{X}^\top - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{X}^\top)^\top (\boldsymbol{X}^\top - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{X}^\top))$$

$$= \|\boldsymbol{X}^\top - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{X}^\top\|_F^2$$

- Here $\|\cdot\|_F$ is the Frobenius matrix norm. The Frobenius norm of an $m \times n$ matrix $\boldsymbol{A} = (a_{ij})$, denoted $\|\cdot\|_F$, is defined by

$$\|\boldsymbol{A}\|_F \equiv \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2} = \sqrt{\text{tr}(\boldsymbol{A}^\top \boldsymbol{A})}$$

Minimization of reconstruction error

- Thus we can rewrite Eq. (2) in a matrix format as

$$\min_{\boldsymbol{B}} \frac{1}{N} \|\boldsymbol{X}^\top - \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{X}^\top\|_F^2 \equiv \frac{1}{N} \left\|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{B}\boldsymbol{B}^\top\right\|_F^2$$

$$\text{subject to } \boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_{M\times M}$$

Minimization of reconstruction error

- Two formulations are equivalent since

reconstruction err. total variance projected variance

$$\frac{1}{N}\|\boldsymbol{X}^\top - \boldsymbol{B}\boldsymbol{B}^\top\boldsymbol{X}^\top\|_F^2 = \frac{1}{N}\operatorname{tr}\left(\boldsymbol{X}^\top\boldsymbol{X}\right) - \operatorname{tr}\left(\boldsymbol{B}^\top\boldsymbol{S}\boldsymbol{B}\right)$$

Proof

$$
\begin{aligned}
\frac{1}{N}\left\|\boldsymbol{X}^{\top}-\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{X}^{\top}\right\|_F^2 &= \frac{1}{N}\left\|\boldsymbol{X}-\boldsymbol{X}\boldsymbol{B}\boldsymbol{B}^{\top}\right\|_F^2 \\
&= \frac{1}{N}\operatorname{tr}\left(\left(\boldsymbol{X}-\boldsymbol{X}\boldsymbol{B}\boldsymbol{B}^{\top}\right)^{\top}\left(\boldsymbol{X}-\boldsymbol{X}\boldsymbol{B}\boldsymbol{B}^{\top}\right)\right) \\
&= \frac{1}{N}\operatorname{tr}\left(\boldsymbol{X}^{\top}\boldsymbol{X}\right)-\frac{2}{N}\operatorname{tr}\left(\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{X}^{\top}\boldsymbol{X}\right) \\
&\quad+\frac{1}{N}\operatorname{tr}\left(\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{X}^{\top}\boldsymbol{X}\boldsymbol{B}\boldsymbol{B}^{\top}\right) \\
&= \frac{1}{N}\operatorname{tr}\left(\boldsymbol{X}^{\top}\boldsymbol{X}\right)-\frac{2}{N}\operatorname{tr}\left(\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{X}^{\top}\boldsymbol{X}\right) \\
&\quad+\frac{1}{N}\operatorname{tr}\left(\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{X}^{\top}\boldsymbol{X}\right) \\
&= \frac{1}{N}\operatorname{tr}\left(\boldsymbol{X}^{\top}\boldsymbol{X}\right)-\frac{1}{N}\operatorname{tr}\left(\boldsymbol{B}\boldsymbol{B}^{\top}\boldsymbol{X}^{\top}\boldsymbol{X}\right) \\
&= \frac{1}{N}\operatorname{tr}\left(\boldsymbol{X}^{\top}\boldsymbol{X}\right)-\operatorname{tr}\left(\boldsymbol{B}^{\top}\boldsymbol{S}\boldsymbol{B}\right)
\end{aligned}
$$